

Differential Geometry

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curvature

Velocity vector

$$T(s) = \gamma'(s)$$

has unit length and is tangent to the curve at point

$$p = \gamma(s)$$

and curvature at p is

$$T'(s) = \gamma''(s)$$

therefore.

signed curvature or simply curvature is

$$\kappa = \langle T', n \rangle = \langle \gamma'', n \rangle$$

and this is how we define it.

Surfaces in Space

Gauss bonnet theorem states that for a oriented surface, gaussian curvature K is a function on a surface

$$\int_M K dS = \int \frac{1}{a^2} dS = \frac{1}{a^2} 4\pi a^2 = 4\pi$$

where for the sphere M of radius a is in this case.

We define $\chi(M)$ as euler characteristic where for previous example it would be 2.

$$\int_M K dS = 2\pi\chi(M)$$

Directional Derivatives

to compute directional derivatives,

$$x^i = p^i + ta^i$$

we set the parametric equations for the line through p in the direction X_p
let

$$a = (a^1, \dots, a^n)$$

then directional derivative $D_{X_p}f$ is

$$\begin{aligned} D_{X_p}f &= \lim_{t \rightarrow 0} \frac{f(p + ta) - f(p)}{t} = \frac{d}{dt} \Big|_{t=0} f(p + ta) \\ &= \Sigma \frac{\partial f}{\partial x^i} * \frac{dx^i}{dt} \Big| = \Sigma \frac{\partial f}{\partial x^i} * a^i \\ &= (\Sigma a^i \frac{\partial}{\partial x^i} \Big|_p) f = X_p f \end{aligned}$$

and therefore directional derivative at p of a C^∞ vector field is

$$D_{X_p}Y = \Sigma (X_p b^i) \frac{\partial}{\partial x^i} \Big|_p$$

The directional Derivative satisfies this property

$$1. D_X Y \text{ is } \mathcal{F}\text{-linear in } Y \text{ and } \mathbb{R} \text{ linear in } X$$

$$2. D_X(fY) = (Xf)Y + fD_X Y$$

we also define

$$[X, Y] = D_X Y - D_Y X$$

and the quantity

$$T(X, Y) = D_X Y - D_Y X - [X, Y]$$

as torsion.

The definition for curvature is

$$R(X, Y) = [D_X, D_Y] - D_{[X, Y]} = D_X D_Y - D_Y D_X - D_{[X, Y]} \in \text{End}_{\mathbb{R}}(X(M))$$

The Shape Operator

for

$$X_p \in T_p M$$

we define shape operator as

$$L_p(X_p) = -D_{X_p} N$$

Lemma let M be the surface in $\mathbb{R}^3$ then

$$\forall X, Y \in \chi(M), \langle L(X), Y \rangle = \langle D_X Y, N \rangle$$

this is because

$$0 = X \langle Y, N \rangle = \langle D_X Y, N \rangle + \langle Y, D_X N \rangle = \langle D_X Y, N \rangle - \langle Y, L(X) \rangle$$

shape operator is self-adjoint as well, meaning

$$\langle L(X), Y \rangle = \langle X, L(Y) \rangle$$

we define second fundamental form as

$$\Pi(X, Y) = \langle L(X), Y \rangle$$

where first fundamental form is euclidean riemannian metric as we've defined earlier.

Affine connection

affine connection is way of differentiating vector fields on manifold, where it is defined as

$$\nabla_{X_p} f = X_p f$$

and to do so, we define in the same way, nabla is defined as

$$\nabla : \chi(M) \times \chi(M) \rightarrow \chi(M)$$

then for

$$\forall X, Y \in \chi(M),$$

it happens the same properties,

$$\nabla_X Y \text{ is } \mathcal{F}\text{-linear in } X$$

$$\nabla_X (fY) = (Xf)Y + f\nabla_X Y$$

it also has same definitions of torsion and curvature, where torsion is in $\chi(M)$ while euclidean connection on $\mathbb{R}^n$ so curvature is in endomorphism or the manifold

for this observation, affine connection ∇ on a manifold M gives linear map

$$\chi(M) \rightarrow \text{End}_{\mathbb{R}}\chi(M)$$

and this is map from

$$X \rightarrow \chi(M)$$

Theorema Egregium

$$i.K_p = \langle R_p(e_1, e_2)e_2, e_1 \rangle$$

ii.gaussian curvature is an isometric invariant of smooth surfaces in $\mathbb{R}^3$

Generalizations to Hypersurfaces

Theorem: let M be a smooth hypersurface and D the directional derivative then

$$\forall X, Y \in \chi(M), \nabla_X Y = (D_X Y)_{tan}$$

where for torsion freeness,

$$D_X Y = \nabla_X Y + (D_X Y)_{normal}$$

and as we define forms of fundamental, first, second and third fundamental form is

$$I(X_p, Y_p) = \langle X_p, Y_p \rangle$$

$$II(X_p, Y_p) = \langle L(X_p), Y_p \rangle = \langle X_p, L(Y_p) \rangle$$

$$III(X_p, Y_p) = \langle L^2(X_p), Y_p \rangle = \langle L(X_p), L(Y_p) \rangle = \langle X_p, L^2(Y_p) \rangle$$

Prop if N is a smooth unit normal vector field on hypersurface M in $\mathbb{R}^{n+1}$ and $X, Y \in \chi(M)$ then

$$(D_X Y)_{nor} = II(X, Y)N$$

connections on vector bundles

connection on a map is defined as

$$\nabla : \chi(M) \times \Gamma(E) \rightarrow \Gamma(E)$$

such that

$$X \in \chi(M), s \in \Gamma(E)$$

then states

i. ∇ is $\mathcal{F}$ -linear in X and $\mathbb{R}$ -linear in s

$$*ii.* \nabla_X(fs) = (Xf)s + f\nabla_X s$$

for connection

$$\nabla^U : \chi(U) \times \Gamma(U, E) \rightarrow \Gamma(U, E)$$

it is defined as

$$\nabla_X e_j = \sum w_j^i(X) e_i$$

where

$$R(X, Y) e_j = \sum \Omega_j^i(X, Y) e_i$$

and

$$R(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}$$

and because of the definition of nabla,

$$R(X, Y) = (Xw_j^i(Y) - Yw_j^i(X) - w_j^i([X, Y]))e_i + (w_k^i(X)w_j^i(Y) - w_k^i(Y)w_j^k(X))e_i$$

$$= dw_j^i(X, Y)e_i + w_k^i \wedge w_j^k(X, Y)e_i$$

therefore we can conclude that

$$\Omega_j^i = dw_j^i + \sum_k w_k^i \wedge w_j^k$$

also called as second structural equation, we also denote that τ 's differential factor is ω , therefore we can also conclude

first structural formula:

$$\tau^i = d\theta^i + \sum_j w_j^i \wedge \theta^j$$

Geodesics

It is called **covariant derivative** if this is allowed

$$\frac{D}{dt} : \Gamma(TM|_{c(t)}) \rightarrow \Gamma(TM|_{c(t)})$$

for this map,

$$\begin{aligned}\frac{DV}{dt} & \text{ is } \mathbb{R} \text{ linear in } V \\ \frac{D(fV)}{dt} &= \frac{df}{dt}V + f\frac{DV}{dt} \\ \frac{DV}{dt}(t) &= \nabla_{c'(t)}V\end{aligned}$$

Christoffel symbol is a function that is of the connection ∇ on the coordinate open set

$$(U, x^1, \dots, x^n)$$

and defined as

$$\nabla_{\partial_i}\partial_j = \sum_{k=1}^n \Gamma_{ij}^k \partial_k$$

and for affine connection ∇ on a manifold is torsion free iff every coordinate chart

$$(U, x^1, \dots, x^n)$$

the Christoffel symbol is symmetric in i and j:

$$\Gamma_{ij}^k = \Gamma_{ji}^k$$

ON a manifold with a connection, $c(t)$ the parametrized curve is a geodesic iff for

$$(U, \phi) = (U, x^1, \dots, x^n)$$

the components of

$$\ddot{y}^k + \sum_{i,j} \Gamma_{ij}^k \dot{y}^i \dot{y}^j = 0$$

this is because

$$\begin{aligned}T(t) &= c'(t) = \sum \dot{y}^j \partial_{j,c(t)} \\ \frac{DT}{dt}(t) &= \sum_j \ddot{y}^j \partial_{j,c(t)} + \sum_j \dot{y}^j \nabla_{c'(t)} \partial_j\end{aligned}$$

$$= \sum_k (\ddot{y}^k + \sum_{i,j} \dot{y}^i \dot{y}^j \Gamma_{ij}^k) \partial_k$$

Gauss-bonnet theorem for surface

the euler characteristic is defined as

$$\chi = V - E + F$$

where we concluded

$$\int_M K vol = 2\pi\chi(M)$$